

SA2 Mock Examination

Lectures 17–29 · Exam 2 / FA-SA2 Scope

77 Questions · Single Best Answer · Weighted by Official Lecture Hours

This mock exam distributes 77 questions across Lectures 17–29 in exact proportion to the official 15-hour schedule for this half of the course (see the distribution table on the next page). Every question carries a single best answer and five options.

Sourcing. The large majority of items are pulled directly from the fully re-verified, 225-question Lec17–29 past-paper key, where every answer has already been checked against source neuroscience material (not the original error-prone colour-detection extraction). Lecture 29 (Clinical Correlations) only had a single genuine past-paper item, so its remaining slots were filled two ways: first by reclassifying clinical-vignette questions that had been filed under “Mixed / Other Topics” in the archive but that are, in substance, exactly the kind of integrative localization case Lecture 29 teaches; and second by writing a small number of original questions (marked †) grounded directly in the EP2 “Clinical Correlation and Neuro Exam” lecture slides — covering motor grading, UMN/LMN localization, reflex levels, spinal cord syndromes, and cerebellar/NMJ examination findings. Lecture 25 (ANS II) was one question short of its hour-weighted quota; this was filled with one reclassified item and two original (†) items on standard ANS-II physiology.

Every question in the Answer Key section states exactly where it came from, so you can weight your confidence accordingly — verified-archive answers first, and †-marked original questions treated as high-yield practice rather than confirmed past-paper fact.

How to use this booklet

1. Work through the Questions section (pp. 3–...) first, lecture by lecture, without looking ahead.
2. Grade yourself against the Answer Key & Explanations section, which mirrors the question order.
3. Read the **Common trap** line for every question you miss — in this exam bank, the wrong options are deliberately built around the same mix-ups that show up on the real SA2, so the trap note is often more useful than the explanation itself.
4. Questions marked † are original (not verified past-paper) — treat them as high-confidence practice grounded in lecture content, not as guaranteed past-exam phrasing.

Question Distribution by Lecture Hours

Lecture	Hours	Questions	% of Exam
Lecture 17 — Motor System	1.0 hr	5	6.5%
Lecture 18 — Cerebellum	1.0 hr	5	6.5%
Lecture 19 — Cerebral Cortex	1.5 hr	8	10.4%
Lecture 20 — Basal Ganglia	1.0 hr	5	6.5%
Lecture 21 — Hypothalamus & Epithalamus	0.5 hr	3	3.9%
Lecture 22 — Thalamus	1.0 hr	5	6.5%
Lecture 23 — Limbic System & Olfaction	0.5 hr	3	3.9%
Lecture 24 — Autonomic Nervous System I (Structures)	1.0 hr	5	6.5%
Lecture 25 — Autonomic Nervous System II (Functions)	1.0 hr	5	6.5%
Lecture 26 — EEG & Sleep Physiology	1.0 hr	5	6.5%
Lecture 27 — Reflex Mechanism	1.0 hr	5	6.5%
Lecture 28 — Blood Supply of Brain & Meninges	1.5 hr	8	10.4%
Lecture 29 — Clinical Correlations	3.0 hr	15	19.5%
Total	15.0 hr	77	100%

Lecture 29 carries the largest single share (15 questions, 19.5%) because the official schedule allocates it 3.0 of the 15 total hours for this half of the course — more than any other single lecture. Its content here spans classic clinical-localization vignettes (cranial nerve palsies, Horner's syndrome, internal capsule syndromes, PCA/midbrain syndromes) plus original questions on physical-exam method (reflex grading, UMN/LMN signs, spinal cord syndromes, cerebellar and NMJ examination) drawn from the EP2 Neuro Exam lecture.

Section 1 — Questions (77 items, single best answer)

Items marked † are original questions written for this mock exam, grounded in lecture content but not verified past-paper items.

Lecture 17 — Motor System — 1.0 hr — 5 questions

1. Regarding the myotatic (knee-jerk) reflex, which of the following is correct?

- A. Afferent = Ib fiber
- B. Efferent = Ia fiber
- C. Antagonist = quadriceps
- D. Heteronymous = hamstring
- E. Receptor = Golgi tendon organ

2. Which statement about muscle spindle intrafusal fibers is correct?

- A. Bag: static; Chain: static
- B. Bag: static & dynamic; Chain: static
- C. Bag: dynamic; Chain: dynamic
- D. Bag: static; Chain: static & dynamic
- E. Bag: static & dynamic; Chain: dynamic

3. A patient has left lower facial weakness and the tongue deviates to the left on protrusion. Where is the lesion?

- A. Right corticospinal tract
- B. Right corticobulbar tract
- C. Left facial motor nucleus + hypoglossal nucleus
- D. Left corticobulbar tract + left hypoglossal nucleus
- E. Left corticospinal tract

4. Which of the following is NOT a pathological reflex?

- A. Babinski sign
- B. Clonus
- C. Cremasteric reflex
- D. Hoffman sign
- E. Hyperreflexia

5. A patient has had a left pyramidal tract lesion for one month. Which finding would NOT be expected?

- A. Hypertonia
- B. Hyperreflexia
- C. Paresis/paralysis
- D. Muscular atrophy
- E. Babinski sign

Lecture 18 — Cerebellum — 1.0 hr — 5 questions

6. The output neuron of the cerebellar cortex is the:

- A. Golgi cell
- B. Granule cell
- C. Basket cell
- D. Stellate cell
- E. Purkinje cell

7. Which of the following tracts to the cerebellum does NOT carry proprioceptive information?

- A. Dorsal spinocerebellar
- B. Ventral spinocerebellar
- C. Olivocerebellar
- D. Tectocerebellar
- E. Vestibulocerebellar

8. Fibers from the dentate nucleus of the cerebellum travel to the ____ of the thalamus by passing through the ____.

- A. VA / Superior cerebellar peduncle
- B. VA / Inferior cerebellar peduncle
- C. VL / Superior cerebellar peduncle
- D. VL / Inferior cerebellar peduncle
- E. VPL / Middle cerebellar peduncle

9. The AICA (Anterior Inferior Cerebellar Artery) supplies all of the following EXCEPT:

- A. Dorsolateral caudal pons
- B. Inferior cerebellar peduncle
- C. Middle cerebellar peduncle
- D. Crus cerebri
- E. Anterior surface of cerebellum

10. Climbing fibers to the cerebellum originate from the:

- A. Red nucleus
- B. Inferior olivary nucleus
- C. Vestibular nuclei
- D. Pontine nuclei
- E. Reticular formation

Lecture 19 — Cerebral Cortex — 1.5 hr — 8 questions

11. A 73-year-old woman has right arm weakness and left gaze preference after a stroke. Sensory exam and pupillary light reflexes are normal. Which lobe is most likely involved?

- A. Frontal
- B. Parietal
- C. Limbic
- D. Occipital
- E. Temporal

12. Which of the following is NOT a feature of Klüver-Bucy syndrome?

- A. Lesions of the anterior thalamus
- B. Hypersexuality
- C. Hyperorality
- D. Psychic blindness
- E. Bilateral temporal lobe lesion

13. A 63-year-old woman has spastic paralysis, right hemianesthesia, and right facial droop but no aphasia. The lesion is in the:

- A. Right cortex
- B. Base of pons (left)
- C. Crus cerebri (left)
- D. Pyramid of medulla (right)
- E. Posterior limb & genu of left internal capsule

14. Ischemia of the paracentral lobule and cingulate gyrus results from occlusion of the:

- A. Frontopolar artery
- B. Callosomarginal artery
- C. Precentral artery
- D. Paracentral artery
- E. Calcarine artery

15. A patient has finger agnosia and dyscalculia. The lesion is in the:

- A. Left angular gyrus
- B. Right angular gyrus
- C. Left precuneus
- D. Right precuneus
- E. Left supramarginal gyrus

16. An MRI shows a lesion in the right pars opercularis. The patient will likely have what disorder?

- A. Motor apraxia
- B. Motor aphasia
- C. Dysarthria
- D. Right hemiplegia
- E. Dysprosody

17. The fiber tract connecting Broca's and Wernicke's areas is the:

- A. Corpus callosum
- B. Internal capsule
- C. Arcuate fasciculus
- D. Superior longitudinal fasciculus
- E. Uncinate fasciculus

18. Alexia without agraphia (pure word blindness) is caused by a lesion in the:

- A. Broca's area
- B. Angular gyrus
- C. Splenium of the corpus callosum
- D. Wernicke's area
- E. Precentral gyrus

Lecture 20 — Basal Ganglia — 1.0 hr — 5 questions

19. Which connection in the basal ganglia circuit is excitatory?

- A. GPe → STN
- B. GPi → VL thalamus
- C. STN → GPi
- D. Striatum → GPi
- E. Striatum → GPe

20. Which basal ganglia structure is in the midbrain?

- A. Caudate
- B. Putamen
- C. Subthalamic nucleus
- D. Globus pallidus externus
- E. Substantia nigra (pars reticulata)

21. A patient with uncontrolled movements has atrophy of the basal ganglia. Her mother had similar symptoms. What is her condition?

- A. Bradykinesia
- B. Parkinson Disease
- C. Hypokinetic disorder
- D. Huntington's Disease
- E. Sydenham's chorea

22. A patient has Parkinson's disease. What is the effect on the basal ganglia pathways?

- A. Overactivity of the direct pathway, underactivity of the indirect pathway
- B. Underactivity of the direct pathway, overactivity of the indirect pathway
- C. Increased excitatory effect of the direct pathway
- D. Increased inhibition of the indirect pathway
- E. No change in either pathway

23. Which supraspinal site is the key initiator of descending analgesia recruited by opioids?

- A. Solitary nucleus
- B. Substantia nigra
- C. Periaqueductal gray (PAG)
- D. Ventral posterior thalamus
- E. Primary motor cortex

Lecture 21 — Hypothalamus & Epithalamus — 0.5 hr — 3 questions

24. A 30-year-old man had mammillary body damage. Which symptom is found?

- A. Hyperorality
- B. Hypersexuality
- C. Facial blunting
- D. Loss of recent memory
- E. Loss of long-term memory

25. Jet lag is due to dysregulation of the:

- A. Pineal gland
- B. Suprachiasmatic nucleus
- C. Paraventricular nucleus
- D. Supraoptic nucleus
- E. Locus coeruleus

26. The heat gain center in the hypothalamus is located in the:

- A. Anterior nucleus
- B. Posterior nucleus
- C. Arcuate nucleus
- D. Suprachiasmatic nucleus
- E. Lateral hypothalamus

Lecture 22 — Thalamus — 1.0 hr — 5 questions

27. Which thalamic nucleus receives input from the contralateral lateral spinothalamic tract?

- A. Centromedian nucleus
- B. Mediodorsal nucleus
- C. Ventral anterior nucleus
- D. Ventral lateral nucleus
- E. Ventral posterolateral nucleus

28. Which sensation bypasses the thalamus before reaching the cortex?

- A. Vision
- B. Hearing
- C. Fine touch
- D. Pain
- E. Olfaction

29. Which thalamic nucleus connects to the solitariothalamic tract?

- A. Centromedian nucleus
- B. Mediodorsal nucleus
- C. Ventral anterior nucleus
- D. Ventral lateral nucleus
- E. Ventral posteromedial nucleus

30. Which thalamic nuclei influence activity of the motor cortex?

- A. Dorsomedial & lateral dorsal nuclei
- B. Ventral anterior & ventral lateral nuclei
- C. Lateral posterior & ventral posteromedial nuclei
- D. Intralaminar & midline nuclei
- E. Lateral geniculate & reticular nuclei

31. Within the Papez circuit, damage to which thalamic nucleus impairs recent memory?

- A. VPM
- B. Dorsomedial
- C. Anterior nucleus
- D. Pulvinar
- E. Lateral geniculate

Lecture 23 — Limbic System & Olfaction — 0.5 hr — 3 questions

32. A 39-year-old woman has an eating disorder, thirst, and fluctuating temperature. Which structure is dysfunctional?

- A. Amygdala
- B. Hippocampus
- C. Hypothalamus
- D. Septal nucleus
- E. Habenula

33. The conversion of short-term to long-term memory occurs in the:

- A. Thalamus
- B. Basal ganglia
- C. Hypothalamus
- D. Amygdala
- E. Hippocampus

34. A 65-year-old alcoholic presents with severe memory impairment. Atrophy is expected in which structure?

- A. Cingulate gyrus
- B. Parahippocampal gyrus
- C. Hippocampus
- D. Amygdala
- E. Mammillary body

Lecture 24 — Autonomic Nervous System I (Structures) — 1.0 hr — 5 questions

35. Throughout the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to which receptor?

- A. N2
- B. M3
- C. $\alpha 1$
- D. $\alpha 2$
- E. $\beta 2$

36. A lesion in which location can present with Horner's syndrome?

- A. Ciliary ganglion
- B. Stellate ganglion
- C. Superior cervical ganglion
- D. Middle cervical ganglion
- E. Sympathetic chain T1–T4

37. Which of the following nerves carries parasympathetic fibers?

- A. Lumbar splanchnic
- B. Greater splanchnic
- C. Lesser splanchnic
- D. Pelvic splanchnic
- E. Least splanchnic

38. Which sympathetic/parasympathetic action pairing is correct?

- A. Pupils: Symp constrict, Para dilate
- B. Heart: Symp ↓HR, Para ↑HR
- C. Sweat glands: Symp ↓, Para ↑
- D. Bladder: Symp relaxes, Para contracts
- E. Male organ: Symp erection, Para ejaculation

39. The primary source of circulating epinephrine is:

- A. Postganglionic sympathetic neurons
- B. Preganglionic sympathetic neurons
- C. Adrenal medullary chromaffin cells
- D. Parasympathetic postganglionic neurons
- E. Preganglionic parasympathetic neurons

Lecture 25 — Autonomic Nervous System II (Functions) — 1.0 hr — 5 questions

40. Which enzyme converts norepinephrine to epinephrine in the adrenal medulla?

- A. Choline acetyltransferase
- B. Acetylcholinesterase
- C. Phenylethanolamine N-methyltransferase (PNMT)
- D. Epinephrine synthase
- E. AA-NAT

41. Norepinephrine for arousal and alertness is synthesized mainly in the:

- A. Raphe nuclei
- B. Substantia nigra
- C. Locus coeruleus
- D. Basal forebrain
- E. Hypothalamus

42. Which of the following is a sympathetic effect?

- A. Pupil constriction
- B. Increased heart rate
- C. Increased small intestine contraction
- D. Limb vessel dilation
- E. Increased urination

43. Which autonomic effector organ receives sympathetic innervation that is cholinergic (acting on muscarinic receptors) rather than adrenergic? †

- A. Heart
- B. Sweat glands
- C. Bronchial smooth muscle
- D. Arrector pili muscles
- E. Blood vessels of the gut

44. The widespread, simultaneous activation of the sympathetic nervous system that produces the 'fight-or-flight' response is best described as: †

- A. Referred pain
- B. Autonomic dysreflexia
- C. Cannon's mass discharge
- D. Vasovagal reflex
- E. Enteric reflex

Lecture 26 — EEG & Sleep Physiology — 1.0 hr — 5 questions

45. What is a defining characteristic of the EEG α -rhythm?

- A. Associated with deep sleep
- B. Frequency of 20–30 Hz
- C. Disappears with eye opening
- D. Replaced by slower waves in REM
- E. Appears in normal waking state

46. A 10-year-old boy has a generalized seizure with momentary loss of consciousness. His EEG shows 2.5–3 Hz generalized spike-wave activity. What is the diagnosis?

- A. Clonic
- B. Absence
- C. Tonic-clonic
- D. Simple partial
- E. Complex partial

47. A 35-year-old man has obstructive sleep apnea. Which EEG pattern occurs during REM sleep?

- A. High amplitude & frequency
- B. Low amplitude & frequency
- C. High amplitude, low frequency
- D. Low amplitude, high frequency
- E. K complexes & spindles

48. A 35-year-old man transitions from non-REM sleep to wakefulness. Which neurotransmitter change occurs?

- A. ↓NE, ↑5-HT, ↑ACh, ↓Histamine, ↓GABA
- B. ↓NE, ↑5-HT, ↑ACh, ↓Histamine, ↑GABA
- C. ↓NE, ↓5-HT, ↑ACh, ↑Histamine, ↑GABA
- D. ↑NE, ↑5-HT, ↑ACh, ↑Histamine, ↓GABA
- E. ↑NE, ↓5-HT, ↓ACh, ↑Histamine, ↓GABA

49. Which statement about sleep stages is INCORRECT?

- A. Stage 1: drowsy and eyes close
- B. Stage 2: sleep spindle
- C. Stage 3: delta and theta wave
- D. Stage 4: slow-wave-sleep
- E. REM sleep: night terror

Lecture 27 — Reflex Mechanism — 1.0 hr — 5 questions

50. Which receptor mediates corneal reflex sensation?

- A. Merkel disks
- B. Ruffini endings
- C. Free nerve endings
- D. Pacinian corpuscles
- E. Meissner corpuscles

51. The cause of the rebound phenomenon (a cerebellar sign) is:

- A. No stretch reflex
- B. Postural disturbance
- C. Muscle spasm
- D. Areflexia
- E. Loss of muscle power

52. The Glasgow Coma Scale (GCS) consists of assessing:

- A. Motor, verbal responses, eye opening
- B. Reflexes, orientation, motor responses
- C. Pupillary response, verbal, pain response
- D. Temperature, pulse, blood pressure
- E. Cranial nerve function, strength, coordination

53. In the pupillary light reflex, sensory input is via CN ____ and motor output is via CN ____.

- A. II, III
- B. II, IV
- C. III, V
- D. V, VII
- E. III, VI

54. The gag reflex is mediated by which brainstem level?

- A. Midbrain
- B. Pons
- C. Medulla oblongata
- D. Diencephalon
- E. Cervical cord

Lecture 28 — Blood Supply of Brain & Meninges — 1.5 hr — 8 questions

55. A patient has suffered a stroke and has difficulty comprehending spoken language (aphasia). Branches of which artery are compromised?

- A. ACA
- B. MCA
- C. PCA
- D. Striate artery
- E. Anterior choroidal artery

56. The superior sagittal sinus is located between:

- A. Inner table of skull & endosteal dura
- B. Endosteal dura & meningeal dura
- C. Meningeal dura & arachnoid
- D. Arachnoid & pia
- E. Pia mater & arachnoid mater

57. An occlusion of the right ACA (anterior cerebral artery) will cause deficits in:

- A. Motor & sensory in right lower limbs
- B. Motor & sensory in left lower limbs
- C. Motor & sensory in right upper limbs
- D. Motor & sensory in left upper limbs
- E. Motor & sensory in all limbs

58. A 46-year-old man presents with apathy, severe right leg weakness greater than arm/face weakness, and urinary incontinence. Which artery is mainly occluded?

- A. Lateral striate artery
- B. Right MCA
- C. Left MCA
- D. Right ACA
- E. Left ACA

59. A pituitary tumor typically causes which visual defect?

- A. Optic nerve lesion
- B. Optic chiasm lesion
- C. Optic tract lesion
- D. Parietal upper optic radiation lesion
- E. Temporal lower optic radiation lesion

60. Which of the following is NOT a criterion used in the ABCD2 score for stroke risk after a TIA?

- A. Hyperlipidemia
- B. Diabetes
- C. Age
- D. Clinical feature
- E. Blood pressure (Hypertension)

61. An embolus from the internal carotid artery is most likely to lodge in which branch?

- A. Middle cerebral artery (MCA)
- B. Posterior communicating artery (PcomA)
- C. Anterior cerebral artery (ACA)
- D. Anterior communicating artery (AcomA)
- E. Posterior cerebral artery (PCA)

62. The arterial supply of the primary visual cortex is the:

- A. Middle cerebral artery
- B. Anterior cerebral artery
- C. Calcarine artery (PCA branch)
- D. Superior cerebellar artery
- E. Anterior choroidal artery

Lecture 29 — Clinical Correlations — 3.0 hr — 15 questions

63. A 37-year-old woman presents with fatigue, difficulty speaking, ptosis, and diplopia that worsens throughout the day. Sensory exam is normal. What is the diagnosis?

- A. Multiple sclerosis
- B. Guillain-Barré syndrome
- C. Lambert-Eaton Syndrome
- D. Tetanus
- E. Myasthenia Gravis

64. A 50-year-old man has ptosis, anhidrosis, and miosis due to a lung apex (Pancoast) tumor. Which muscle is responsible for the drooping eyelid?

- A. Ciliary muscle
- B. Pupillary sphincter
- C. Pupillary dilator
- D. Superior tarsal muscle
- E. Arrector pili

65. The same 50-year-old man with a lung apex tumor also has miosis. Which muscle correlates with this finding?

- A. Ciliary muscle
- B. Arrector pili
- C. Pupillary dilator
- D. Superior tarsal
- E. Hamstring

66. A 30-year-old woman with a UTI has urinary urgency. Which receptor mediates detrusor muscle contraction during voiding?

- A. β_2
- B. β_3
- C. N
- D. M2
- E. M3

67. A lesion in the left cerebral cortex affects the corticobulbar fibers. Which finding is true for this lesion?

- A. Right ptosis
- B. Right inability to close the eyelid
- C. Right weakness closing the mouth tightly
- D. Left inability to raise the shoulder
- E. Right tongue deviation

68. The mechanism of action for diazepam (a benzodiazepine) used in status epilepticus and anxiety disorders is:

- A. Blocks GABA reuptake
- B. Increases the duration of Cl⁻ channel opening
- C. Increases the frequency of Cl⁻ channel opening at the GABA-A receptor
- D. Blocks NMDA receptor Ca²⁺ influx
- E. Enhances glycine binding

69. A patient has vertical diplopia and difficulty looking down with the left eye, especially when the eye is adducted. What is the cranial nerve lesion?

- A. Left CN IV
- B. Left CN III
- C. Right CN II
- D. Left CN VI
- E. Right CN VI

70. A patient has contralateral sensory loss and ipsilateral CN III palsy. This combination is caused by occlusion of the:

- A. ACA
- B. MCA
- C. PCA
- D. Anterior choroidal artery
- E. Basilar artery

71. Using the standard mnemonic for deep tendon reflex root levels (counting upward from the ankle: 1-2, 3-4, 5-6, 7-8), which nerve root level corresponds to the triceps reflex? †

- A. S1, S2
- B. L3, L4
- C. C5, C6
- D. C6 only
- E. C7, C8

72. Which clinical finding, if present, favors a lower motor neuron (LMN) lesion over an upper motor neuron (UMN) lesion? †

- A. Increased tone
- B. Increased deep tendon reflexes
- C. Extensor plantar (Babinski) response
- D. Fasciculations
- E. Sustained clonus

73. A patient has purely motor weakness with no sensory complaints. Examination shows fatigability that worsens through the day, but deep tendon reflexes remain preserved. This pattern best localizes to the level of the: †

- A. Anterior horn cell
- B. Nerve root
- C. Peripheral nerve
- D. Neuromuscular junction
- E. Muscle

74. A patient develops sudden bilateral leg weakness and loss of pain/temperature sensation below the umbilicus after an episode of severe hypotension, but vibration and proprioception remain intact. This is most consistent with: †

- A. Brown-Séquard syndrome
- B. Anterior cord syndrome
- C. Central cord syndrome
- D. Subacute combined degeneration
- E. Complete cord transection

75. A patient asked to perform rapid alternating pronation-supination of the forearm performs the movements slowly and irregularly. This finding is best termed: †

- A. Dysmetria
- B. Dyssynergia
- C. Dysdiadochokinesia
- D. Intention tremor
- E. Nystagmus

76. A large cerebellopontine (CP) angle tumor classically produces a nystagmus with a large-amplitude, slow component when looking toward the side of the lesion and a smaller, fast component when looking away. This is known as: †

- A. Downbeat nystagmus
- B. Bruns' nystagmus
- C. See-saw nystagmus
- D. Optokinetic nystagmus
- E. Gaze-evoked nystagmus (physiologic)

77. Which clinical feature helps distinguish botulism from myasthenia gravis? †

- A. Both cause purely motor weakness without sensory loss
- B. Only botulism can cause ptosis and diplopia
- C. Botulism characteristically also involves autonomic dysfunction, while myasthenia gravis does not
- D. Myasthenia gravis is a presynaptic NMJ disorder, while botulism is postsynaptic
- E. Reflexes are typically preserved in botulism but lost in myasthenia gravis

Section 2 — Answer Key & Explanations

Every entry states the correct answer, why it's correct, the specific wrong-answer trap built into the distractors, and where the question came from.

Lecture 17 — Motor System — 1.0 hr — 5 questions

Q1. Regarding the myotatic (knee-jerk) reflex, which of the following is correct?

Correct answer: D. Heteronymous = hamstring

The knee-jerk (myotatic/stretch) reflex is monosynaptic: Ia afferents from the quadriceps muscle spindle excite quadriceps alpha motor neurons directly (homonymous excitation) while simultaneously inhibiting, via an inhibitory interneuron, the antagonist hamstring motor neurons — this antagonist inhibition is the heteronymous component, targeting the hamstring.

Common trap: Options A and E swap Ia and Ib — Ia (muscle spindle) is the stretch-reflex afferent; Ib (Golgi tendon organ) instead mediates the inverse myotatic (autogenic inhibition) reflex, a completely different circuit.

Verified — corrected past-paper key (Q1)

Q2. Which statement about muscle spindle intrafusal fibers is correct?

Correct answer: B. Bag: static & dynamic; Chain: static

Nuclear bag fibers carry both static and dynamic responses (a dynamic bag1 fiber plus a static bag2 fiber); nuclear chain fibers carry only the static response.

Common trap: Any option that gives chain fibers a “dynamic” role inverts the physiology — dynamic velocity-sensitivity is a bag-fiber-only property.

Verified — corrected past-paper key (Q3)

Q3. A patient has left lower facial weakness and the tongue deviates to the left on protrusion. Where is the lesion?

Correct answer: B. Right corticobulbar tract

Lower facial weakness (forehead spared) plus tongue deviation to the same side both reflect a single UMN (corticobulbar) lesion contralateral to the deficits — here, the right corticobulbar tract, which supplies both the contralateral CN VII and CN XII nuclei above their crossing points.

Common trap: Combining two left-sided nuclear lesions (C/D) would require independent damage to two separate brainstem nuclei and would not produce the clean “forehead-sparing” UMN pattern described here.

Verified — corrected past-paper key (Q11)

Q4. Which of the following is NOT a pathological reflex?

Correct answer: C. Cremasteric reflex

The cremasteric reflex is a normal superficial (cutaneous) reflex present in healthy individuals. Babinski sign, clonus, Hoffmann sign, and hyperreflexia are all abnormal UMN findings.

Common trap: Babinski sign is the most “famous” abnormal reflex, which tempts test-takers to mark it — but re-read the stem: it asks which is NOT pathological.

Verified — corrected past-paper key (Q16)

Q5. A patient has had a left pyramidal tract lesion for one month. Which finding would NOT be expected?

Correct answer: D. Muscular atrophy

At one month, the fully chronic UMN pattern (spastic hypertonia, hyperreflexia, paresis, extensor plantar response) is expected. Significant muscular atrophy is a LMN finding (denervation atrophy), not a UMN one — UMN lesions show at most mild disuse atrophy.

Common trap: Students often equate “long-standing weakness” with atrophy regardless of UMN/LMN distinction; true wasting requires loss of the motor unit itself, which only happens with LMN damage.

Verified — corrected past-paper key (Q17)

Lecture 18 — Cerebellum — 1.0 hr — 5 questions

Q6. The output neuron of the cerebellar cortex is the:

Correct answer: E. Purkinje cell

Purkinje cells are the sole output neurons of the cerebellar cortex; they inhibit (via GABA) the deep cerebellar nuclei, which then provide the net output of the cerebellum to the rest of the brain.

Common trap: Basket and stellate cells are also inhibitory, which tempts confusion, but both are cortical interneurons, not projection/output neurons.

Verified — corrected past-paper key (Q33)

Q7. Which of the following tracts to the cerebellum does NOT carry proprioceptive information?

Correct answer: D. Tectocerebellar

Tectocerebellar fibers relay visual and auditory information from the superior and inferior colliculi; the spinocerebellar tracts, by contrast, are the classic proprioceptive relays into the cerebellum.

Common trap: Olivocerebellar (climbing fiber) input is sometimes assumed to be “purely proprioceptive,” but the clean, unambiguous non-proprioceptive tract here is tectocerebellar, which is visual/auditory in origin.

Verified — corrected past-paper key (Q39)

Q8. Fibers from the dentate nucleus of the cerebellum travel to the ____ of the thalamus by passing through the ____.

Correct answer: C. VL / Superior cerebellar peduncle

Dentate nucleus output ascends via the superior cerebellar peduncle to the contralateral VL nucleus of the thalamus, completing the cerebello-thalamo-cortical loop back to motor cortex.

Common trap: VA is the basal-ganglia-recipient motor thalamic nucleus, not the cerebellar-recipient one — an easy VA/VL mix-up.

Verified — corrected past-paper key (Q42)

Q9. The AICA (Anterior Inferior Cerebellar Artery) supplies all of the following EXCEPT:

Correct answer: D. Crus cerebri

The crus cerebri is a midbrain structure, supplied instead by branches of the posterior cerebral artery / basilar perforators. AICA's territory (classic lateral pontine syndrome) is confined entirely to the pontine and cerebellar levels.

Common trap: It is easy to forget that AICA never reaches the midbrain at all, leading to the accidental selection of a genuine pontine AICA target instead of the true outlier.

Verified — corrected past-paper key (Q45)

Q10. Climbing fibers to the cerebellum originate from the:

Correct answer: B. Inferior olivary nucleus

Climbing fibers arise exclusively from the contralateral inferior olivary nucleus and form a strong, one-to-one excitatory synapse onto Purkinje cells.

Common trap: *Pontine nuclei is the classic distractor since it gives rise to the OTHER major cerebellar afferent type — mossy fibers, entering via the middle cerebellar peduncle — not climbing fibers.*

Verified — corrected past-paper key (Q49)

Lecture 19 — Cerebral Cortex — 1.5 hr — 8 questions

Q11. A 73-year-old woman has right arm weakness and left gaze preference after a stroke. Sensory exam and pupillary light reflexes are normal. Which lobe is most likely involved?

Correct answer: A. Frontal

Right arm weakness (left corticospinal tract) plus left gaze preference (unopposed left frontal eye field driving gaze toward the side of the lesion) both localize to the left frontal lobe.

Common trap: *Gaze preference is sometimes wrongly assumed to be a parietal or occipital (visual) sign, but voluntary conjugate gaze is driven by the frontal eye field, not the visual cortex.*

Verified — corrected past-paper key (Q50)

Q12. Which of the following is NOT a feature of Klüver-Bucy syndrome?

Correct answer: A. Lesions of the anterior thalamus

Klüver-Bucy syndrome results from bilateral temporal lobe (amygdala) damage and features hyperorality, hypersexuality, and psychic blindness (visual agnosia). Anterior thalamic lesions are unrelated — those instead cause Papez-circuit amnesia.

Common trap: *Because “bilateral temporal lobe lesion” (E) is the actual cause of the syndrome, some students mistakenly flag it as the “odd one out” instead of spotting the truly unrelated structure.*

Verified — corrected past-paper key (Q54)

Q13. A 63-year-old woman has spastic paralysis, right hemianesthesia, and right facial droop but no aphasia. The lesion is in the:

Correct answer: E. Posterior limb & genu of left internal capsule

Combined contralateral weakness, hemianesthesia, and facial droop without aphasia is the signature of a compact subcortical lesion that bundles motor, sensory, and corticobulbar fibers together before they fan out cortically — the posterior limb and genu of the left internal capsule.

Common trap: *A pontine or medullary lesion (B–D) could cause crossed or purely motor findings, but would not cleanly explain the combined sensory + motor + facial triad, and option D is also mislateralized.*

Verified — corrected past-paper key (Q57)

Q14. Ischemia of the paracentral lobule and cingulate gyrus results from occlusion of the:

Correct answer: B. Callosomarginal artery

The callosomarginal artery, a branch of the ACA, supplies both the paracentral lobule and the cingulate gyrus together.

Common trap: "Paracentral artery" (D) sounds like the obvious match by name alone, but it is a separate, more distal ACA branch supplying only the paracentral lobule proper, not the cingulate gyrus as well.

Verified — corrected past-paper key (Q64)

Q15. A patient has finger agnosia and dyscalculia. The lesion is in the:

Correct answer: A. Left angular gyrus

Finger agnosia and dyscalculia are two components of the Gerstmann tetrad (with agraphia and left-right disorientation), localizing to the dominant (left) angular gyrus.

Common trap: The precuneus is a different parietal structure tied to visuospatial self-processing and is a plausible-sounding but unrelated distractor here.

Verified — corrected past-paper key (Q69)

Q16. An MRI shows a lesion in the right pars opercularis. The patient will likely have what disorder?

Correct answer: E. Dysprosody

The right pars opercularis is the non-dominant-hemisphere homolog of Broca's area; damage there causes dysprosody (aprosodia) — loss of the emotional, melodic tone of speech — mirroring how a left-sided lesion produces true Broca's aphasia.

Common trap: "Motor aphasia" (B) is tempting if the test-taker forgets this is the RIGHT (non-dominant) hemisphere, where classical language function is not localized.

Verified — corrected past-paper key (Q71)

Q17. The fiber tract connecting Broca's and Wernicke's areas is the:

Correct answer: C. Arcuate fasciculus

The arcuate fasciculus is the classic association fiber bundle linking Wernicke's (posterior) and Broca's (anterior) language areas; its damage produces conduction aphasia (fluent speech, poor repetition).

Common trap: The superior longitudinal fasciculus (D) anatomically overlaps with the arcuate fasciculus and is a common near-miss distractor.

Verified — corrected past-paper key (Q78)

Q18. Alexia without agraphia (pure word blindness) is caused by a lesion in the:

Correct answer: C. Splenium of the corpus callosum

Alexia without agraphia results from a splenium of the corpus callosum lesion (typically with a left occipital lesion), disconnecting the intact right visual cortex from left-hemisphere language areas — the patient can still write but cannot read even their own writing back.

Common trap: Angular gyrus (B) lesions instead cause alexia WITH agraphia — an easy pairing to confuse with the splenium syndrome.

Verified — corrected past-paper key (Q80)

Lecture 20 — Basal Ganglia — 1.0 hr — 5 questions

Q19. Which connection in the basal ganglia circuit is excitatory?

Correct answer: C. STN → GPi

The subthalamic nucleus (STN) is the single excitatory (glutamatergic) relay in the basal ganglia circuit — STN → GPi excites the output nucleus, while every other listed connection is inhibitory (GABAergic).

Common trap: All the striatal and pallidal connections look structurally similar, but only the STN uses an excitatory transmitter — memorize STN as the sole exception.

Verified — corrected past-paper key (Q82)

Q20. Which basal ganglia structure is in the midbrain?

Correct answer: E. Substantia nigra (pars reticulata)

The substantia nigra (both pars compacta and pars reticulata) is anatomically a midbrain structure, even though functionally it belongs to the basal ganglia circuit; caudate, putamen, GPe, and STN are all telencephalic or diencephalic.

Common trap: STN is a close runner-up (diencephalic, adjacent to the midbrain) but is not itself a midbrain structure — precise anatomy matters here.

Verified — corrected past-paper key (Q85)

Q21. A patient with uncontrolled movements has atrophy of the basal ganglia. Her mother had similar symptoms. What is her condition?

Correct answer: D. Huntington's Disease

Chorea, caudate/basal ganglia atrophy, and an autosomal-dominant family history together are classic for Huntington's disease (CAG-repeat expansion in the HTT gene).

Common trap: Sydenham's chorea (E) is also choreiform, but is post-streptococcal and self-limited in children, without a hereditary pattern — the family history is the key discriminator.

Verified — corrected past-paper key (Q88)

Q22. A patient has Parkinson's disease. What is the effect on the basal ganglia pathways?

Correct answer: B. Underactivity of the direct pathway, overactivity of the indirect pathway

Dopamine loss removes D1-mediated facilitation of the direct pathway (making it underactive) and removes D2-mediated inhibition of the indirect pathway (making it overactive/disinhibited) — the net effect is increased thalamic inhibition and bradykinesia.

Common trap: It is easy to reverse the direct vs. indirect pathway direction under exam pressure — remember “direct pathway = go, and dopamine loss takes away the go.”

Verified — corrected past-paper key (Q89)

Q23. Which supraspinal site is the key initiator of descending analgesia recruited by opioids?

Correct answer: C. Periaqueductal gray (PAG)

The periaqueductal gray (PAG) is the classic midbrain site that initiates descending analgesia, projecting via the rostral ventromedial medulla down to the dorsal horn.

Common trap: The solitary nucleus (A) is tempting because it is also a brainstem “gray” involved in visceral/autonomic reflexes, but it has no role in descending pain modulation.

Verified — corrected past-paper key (Q91)

Lecture 21 — Hypothalamus & Epithalamus — 0.5 hr — 3 questions

Q24. A 30-year-old man had mammillary body damage. Which symptom is found?

Correct answer: D. Loss of recent memory

The mammillary bodies are part of the Papez circuit (classically damaged in thiamine-deficient Wernicke-Korsakoff syndrome); damage causes anterograde amnesia — loss of the ability to form new (recent) memories.

Common trap: Hyperorality/hypersexuality (A/B) belong to Klüver-Bucy syndrome (temporal lobe/amygdala), an entirely different structure — don't let the "memory-related structure" cue pull you toward temporal-lobe answers.

Verified — corrected past-paper key (Q92)

Q25. Jet lag is due to dysregulation of the:

Correct answer: B. Suprachiasmatic nucleus

Jet lag results from the suprachiasmatic nucleus (the master circadian pacemaker) being out of phase with the new external light-dark cycle.

Common trap: The pineal gland (A) executes the melatonin signal but is not itself the "clock" — the SCN is the actual pacemaker being asked about.

Verified — corrected past-paper key (Q98)

Q26. The heat gain center in the hypothalamus is located in the:

Correct answer: B. Posterior nucleus

The posterior hypothalamic nucleus is the heat-gain/conservation center (mediating shivering and vasoconstriction); the anterior hypothalamic nucleus is instead the heat-loss center (sweating, vasodilation).

Common trap: Many students memorize "anterior = hot, posterior = cold" backwards — try "A for Air-conditioning" (heat loss) as a quick check.

Verified — corrected past-paper key (Q101)

Lecture 22 — Thalamus — 1.0 hr — 5 questions

Q27. Which thalamic nucleus receives input from the contralateral lateral spinothalamic tract?

Correct answer: E. Ventral posterolateral nucleus

VPL is the thalamic relay for body pain, temperature, and touch, receiving the contralateral spinothalamic tract as well as the dorsal column-medial lemniscus pathway.

Common trap: Mediodorsal nucleus (B) is a common distractor because it sounds "sensory," but it actually relays limbic/prefrontal information, not somatosensory input.

Verified — corrected past-paper key (Q102)

Q28. Which sensation bypasses the thalamus before reaching the cortex?

Correct answer: E. Olfaction

Olfaction is the one sensory modality whose primary pathway (lateral olfactory stria to piriform cortex) bypasses the thalamus entirely before reaching cortex — every other modality synapses in a dedicated thalamic relay nucleus.

Common trap: Vision is sometimes mistakenly picked since retinal processing "feels" direct, but vision clearly relays through the lateral geniculate nucleus of the thalamus.

Verified — corrected past-paper key (Q103)

Q29. Which thalamic nucleus connects to the solitariothalamic tract?

Correct answer: E. Ventral posteromedial nucleus

VPM receives the solitariothalamic tract, which carries taste (and general facial sensation) information, and relays it onward to the gustatory cortex (insula/frontal operculum).

Common trap: VPL vs. VPM is a classic mix-up — remember “M for Mouth” (face/taste = VPM) versus VPL for the body.

Verified — corrected past-paper key (Q105)

Q30. Which thalamic nuclei influence activity of the motor cortex?

Correct answer: B. Ventral anterior & ventral lateral nuclei

VA and VL are the thalamic motor relay nuclei — VA receives basal ganglia output and VL receives cerebellar (dentate) output — both projecting onward to motor and premotor cortex.

Common trap: Lateral geniculate (E) is a tempting-sounding “lateral” nucleus, but it is purely visual and has no connection to motor cortex.

Verified — corrected past-paper key (Q106)

Q31. Within the Papez circuit, damage to which thalamic nucleus impairs recent memory?

Correct answer: C. Anterior nucleus

The anterior thalamic nucleus, which receives the mammillothalamic tract and projects onward to the cingulate gyrus, is the key Papez-circuit relay whose damage impairs memory consolidation.

Common trap: The dorsomedial nucleus (B) belongs to a different limbic/prefrontal loop and is a frequent mix-up with the anterior nucleus in Papez-circuit questions.

Verified — corrected past-paper key (Q114)

Lecture 23 — Limbic System & Olfaction — 0.5 hr — 3 questions

Q32. A 39-year-old woman has an eating disorder, thirst, and fluctuating temperature. Which structure is dysfunctional?

Correct answer: C. Hypothalamus

Eating behavior, thirst, and temperature regulation are all classic hypothalamic homeostatic functions; dysfunction across all three together points squarely to the hypothalamus.

Common trap: Hippocampus is a tempting distractor purely because it's a “famous” limbic structure, but it has no role in appetite, thirst, or thermoregulation — those are hypothalamic, not hippocampal, functions.

Verified — corrected past-paper key (Q117)

Q33. The conversion of short-term to long-term memory occurs in the:

Correct answer: E. Hippocampus

The hippocampus is essential for consolidating short-term memories into stable long-term memory.

Common trap: The amygdala (D) is involved in tagging memories with emotional salience but is not the primary consolidation site.

Verified — corrected past-paper key (Q120)

Q34. A 65-year-old alcoholic presents with severe memory impairment. Atrophy is expected in which structure?

Correct answer: E. Mammillary body

Chronic alcoholism causes thiamine deficiency, classically producing mammillary body atrophy as part of Wernicke-Korsakoff syndrome, with resulting severe anterograde amnesia.

Common trap: *Hippocampus (C) is a plausible “memory structure” distractor, but the specific, classically-tested lesion site in alcoholic Korsakoff syndrome is the mammillary body.*

Verified — corrected past-paper key (Q122)

Lecture 24 — Autonomic Nervous System I (Structures) — 1.0 hr — 5 questions

Q35. Throughout the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to which receptor?

Correct answer: A. N2

All preganglionic autonomic fibers, both sympathetic and parasympathetic, release acetylcholine acting on nicotinic (N2) receptors at the ganglionic synapse.

Common trap: *M3 (B) is the classic postganglionic parasympathetic receptor — a tempting choice, but it applies to the wrong synapse (postganglionic, not preganglionic).*

Verified — corrected past-paper key (Q128)

Q36. A lesion in which location can present with Horner's syndrome?

Correct answer: E. Sympathetic chain T1–T4

Horner's syndrome (ptosis, miosis, anhidrosis) can result from a lesion anywhere along the three-neuron oculosympathetic pathway, including the preganglionic sympathetic chain from T1–T4 (e.g., from a Pancoast tumor).

Common trap: *The ciliary ganglion (A) carries parasympathetic fibers for pupillary constriction/accommodation, not sympathetic fibers — a lesion there would not cause Horner's.*

Verified — corrected past-paper key (Q130)

Q37. Which of the following nerves carries parasympathetic fibers?

Correct answer: D. Pelvic splanchnic

Of all the named splanchnic nerves, only the pelvic splanchnic nerve (S2–S4) is parasympathetic; the greater, lesser, least, and lumbar splanchnic nerves are all sympathetic.

Common trap: *“Splanchnic” sounds uniformly sympathetic to most students, making it easy to forget the one parasympathetic exception.*

Verified — corrected past-paper key (Q131)

Q38. Which sympathetic/parasympathetic action pairing is correct?

Correct answer: D. Bladder: Symp relaxes, Para contracts

Sympathetic input relaxes the detrusor and contracts the internal urethral sphincter (favoring storage); parasympathetic input contracts the detrusor and relaxes the sphincter (favoring voiding) — option D correctly states the detrusor half of this relationship.

Common trap: *Option E reverses the classic mnemonic “Point and Shoot” (Parasympathetic = erection/Point, Sympathetic = ejaculation/Shoot) — a very commonly memorized-backwards fact.*

Verified — corrected past-paper key (Q134)

Q39. The primary source of circulating epinephrine is:

Correct answer: C. Adrenal medullary chromaffin cells

Adrenal medullary chromaffin cells are a modified sympathetic ganglion, directly innervated by preganglionic sympathetic fibers, and are the primary source of circulating epinephrine.

Common trap: *Postganglionic sympathetic neurons (A) release norepinephrine locally at synapses rather than circulating epinephrine — a key distinction between local neurotransmission and endocrine-like hormone release.*

Verified — corrected past-paper key (Q136)

Lecture 25 — Autonomic Nervous System II (Functions) — 1.0 hr — 5 questions

Q40. Which enzyme converts norepinephrine to epinephrine in the adrenal medulla?

Correct answer: C. Phenylethanolamine N-methyltransferase (PNMT)

PNMT converts norepinephrine to epinephrine in the adrenal medulla — the final step of catecholamine synthesis, present there because high local cortisol exposure induces PNMT expression.

Common trap: *“Epinephrine synthase” (D) sounds plausible but is not a real enzyme name — a classic made-up distractor.*

Verified — corrected past-paper key (Q139)

Q41. Norepinephrine for arousal and alertness is synthesized mainly in the:

Correct answer: C. Locus coeruleus

The locus coeruleus is the brain's primary noradrenergic nucleus, driving CNS arousal and alertness through widespread cortical and subcortical projections.

Common trap: *Raphe nuclei (A) form the analogous serotonergic (not noradrenergic) arousal system — easy to conflate the two major brainstem monoamine centers.*

Verified — corrected past-paper key (Q140)

Q42. Which of the following is a sympathetic effect?

Correct answer: B. Increased heart rate

Increased heart rate (via β_1 receptors) is a classic sympathetic “fight-or-flight” effect; pupil constriction, gut motility, and urination are all parasympathetic effects.

Common trap: *“Limb vessel dilation” (D) sounds parasympathetic/relaxed, but skeletal-muscle vasodilation during exercise is actually mediated by sympathetic cholinergic fibers — a well-known exception, though heart rate remains the cleanest sympathetic answer here.*

Reclassified — corrected past-paper key (Q195), originally in the Mixed/Other Topics bucket; reassigned to Lecture 25 (ANS Functions) to complete the lecture-hour quota.

Q43. Which autonomic effector organ receives sympathetic innervation that is cholinergic (acting on muscarinic receptors) rather than adrenergic?

Correct answer: B. Sweat glands

Sweat glands are the classic exception to the rule that sympathetic postganglionic fibers are adrenergic: the postganglionic fiber is anatomically sympathetic, but it releases acetylcholine acting on muscarinic receptors.

Common trap: *Arrector pili (D) is often confused with sweat glands since both are “skin” targets, but arrector pili remain typically adrenergic (α_1), unlike sweat glands.*

Original — standard ANS physiology within the Lecture 25 (Functions of the ANS) scope. †

Q44. The widespread, simultaneous activation of the sympathetic nervous system that produces the 'fight-or-flight' response is best described as:

Correct answer: C. Cannon's mass discharge

Unlike the more discrete, localized reflexes typical of the parasympathetic system, the sympathetic nervous system is capable of a mass discharge (described by Walter Cannon) — a coordinated, body-wide activation that produces the classic fight-or-flight response.

Common trap: Autonomic dysreflexia (B) is also a dramatic, widespread sympathetic phenomenon, but it is a pathological reflex seen in spinal cord injury above T6, not the normal physiological basis of fight-or-flight.

Original — standard ANS physiology within the Lecture 25 (Functions of the ANS) scope. †

Lecture 26 — EEG & Sleep Physiology — 1.0 hr — 5 questions

Q45. What is a defining characteristic of the EEG α -rhythm?

Correct answer: C. Disappears with eye opening

Alpha rhythm (8–13 Hz) is associated with relaxed wakefulness with the eyes closed, and is classically defined by its disappearance ("alpha blocking") when the eyes open — the single most specific, testable feature of this rhythm.

Common trap: Option E is technically true but nonspecific — alpha appears only during RELAXED wakefulness with eyes closed, and "disappearance on eye opening" is the sharper, more precisely testable hallmark that distinguishes it from other waking rhythms.

Verified — corrected past-paper key (Q142)

Q46. A 10-year-old boy has a generalized seizure with momentary loss of consciousness. His EEG shows 2.5–3 Hz generalized spike-wave activity. What is the diagnosis?

Correct answer: B. Absence

Brief loss of consciousness with the classic 2.5–3 Hz generalized spike-and-wave pattern in a child is the hallmark of absence (petit mal) seizures.

Common trap: Complex partial seizures (E) also impair consciousness, but show focal (not generalized 2.5–3 Hz spike-wave) discharges and typically last longer with post-ictal confusion, unlike the brief "staring spells" of absence seizures.

Verified — corrected past-paper key (Q143)

Q47. A 35-year-old man has obstructive sleep apnea. Which EEG pattern occurs during REM sleep?

Correct answer: D. Low amplitude, high frequency

REM sleep EEG is desynchronized — low amplitude, high frequency — closely resembling the waking EEG, which is why REM is called "paradoxical sleep."

Common trap: K complexes and spindles (E) are Stage 2 NREM sleep features, not REM — a very common stage mix-up.

Verified — corrected past-paper key (Q144)

Q48. A 35-year-old man transitions from non-REM sleep to wakefulness. Which neurotransmitter change occurs?

Correct answer: D. ↑NE, ↑5-HT, ↑ACh, ↑Histamine, ↓GABA

Waking requires increased arousal-promoting transmitters — norepinephrine, serotonin, acetylcholine, and histamine all rise — while the inhibitory, sleep-promoting transmitter GABA falls.

Common trap: It is tempting to assume “waking = more of everything,” but GABA specifically must fall (it is inhibitory and sleep-promoting) — the one down-arrow in an otherwise all-up pattern.

Verified — corrected past-paper key (Q145)

Q49. Which statement about sleep stages is INCORRECT?

Correct answer: E. REM sleep: night terror

Night terrors are a classic NREM (Stage 3/4, slow-wave sleep) parasomnia, not a REM phenomenon; nightmares, a distinct entity, occur during REM instead.

Common trap: Nightmares and night terrors are frequently used interchangeably by students, but they occur in opposite sleep stages (REM vs. deep NREM, respectively) — a frequently tested distinction.

Verified — corrected past-paper key (Q152)

Lecture 27 — Reflex Mechanism — 1.0 hr — 5 questions

Q50. Which receptor mediates corneal reflex sensation?

Correct answer: C. Free nerve endings

The cornea is densely innervated by free nerve endings (via CN V1, the ophthalmic division) and lacks any specialized encapsulated mechanoreceptors.

Common trap: Pacinian corpuscles (D) are the “go-to” mechanoreceptor answer for many touch/pressure questions, but the cornea specifically has no encapsulated receptors at all.

Verified — corrected past-paper key (Q153)

Q51. The cause of the rebound phenomenon (a cerebellar sign) is:

Correct answer: A. No stretch reflex

The rebound phenomenon (Holmes sign) occurs because the cerebellum normally coordinates rapid antagonist-muscle “checking” via the stretch reflex; without this cerebellar braking, the limb overshoots when resistance is suddenly released.

Common trap: “Loss of muscle power” (E) sounds intuitive for any neurological deficit, but the rebound phenomenon is a coordination problem, not a weakness problem — cerebellar lesions do not cause frank paresis.

Verified — corrected past-paper key (Q155)

Q52. The Glasgow Coma Scale (GCS) consists of assessing:

Correct answer: A. Motor, verbal responses, eye opening

The Glasgow Coma Scale assesses three domains: eye opening (max 4), verbal response (max 5), and motor response (max 6), for a maximum total score of 15.

Common trap: Pupillary response (C) is a genuinely important neuro-exam finding, but it is NOT one of the three GCS components — it is tracked separately.

Verified — corrected past-paper key (Q156)

Q53. In the pupillary light reflex, sensory input is via CN ____ and motor output is via CN ____.

Correct answer: A. II, III

The pupillary light reflex has its afferent limb via CN II (optic nerve) and its efferent limb via CN III (oculomotor, through the Edinger-Westphal nucleus and ciliary ganglion).

Common trap: "V, VII" (D) is actually the afferent/efferent pairing for the CORNEAL reflex, not the pupillary light reflex — a classic mix-up between two closely related brainstem reflexes.

Verified — corrected past-paper key (Q157)

Q54. The gag reflex is mediated by which brainstem level?

Correct answer: C. Medulla oblongata

The gag reflex (afferent CN IX, efferent CN X) is mediated at the level of the medulla oblongata, where both nerves' nuclei reside.

Common trap: Pons (B) is a common guess since several cranial nerve reflexes (corneal, jaw jerk) are pontine, but the CN IX and CN X nuclei specifically are medullary.

Verified — corrected past-paper key (Q158)

Lecture 28 — Blood Supply of Brain & Meninges — 1.5 hr — 8 questions

Q55. A patient has suffered a stroke and has difficulty comprehending spoken language (aphasia). Branches of which artery are compromised?

Correct answer: B. MCA

MCA territory includes the dominant-hemisphere language cortex (Broca's and Wernicke's areas); occlusion is the classic vascular cause of aphasia.

Common trap: Anterior choroidal artery is a tempting distractor because it supplies deep structures near the internal capsule/optic tract, but it does not supply the cortical language areas.

Verified — corrected past-paper key (Q159)

Q56. The superior sagittal sinus is located between:

Correct answer: B. Endosteal dura & meningeal dura

The superior sagittal sinus runs within a split of the dura mater itself, sandwiched between the outer (endosteal/periosteal) and inner (meningeal) dural layers.

Common trap: "Meningeal dura & arachnoid" (C) sounds anatomically adjacent, but the subdural space (between dura and arachnoid) contains bridging veins, not the sagittal sinus itself.

Verified — corrected past-paper key (Q164)

Q57. An occlusion of the right ACA (anterior cerebral artery) will cause deficits in:

Correct answer: B. Motor & sensory in left lower limbs

The ACA supplies the medial cortical surface, including the leg/foot motor and sensory homunculus; a right ACA occlusion therefore causes contralateral (left) leg deficits.

Common trap: It is easy to forget the contralateral rule under exam pressure and pick the ipsilateral (right leg) option instead.

Verified — corrected past-paper key (Q168)

Q58. A 46-year-old man presents with apathy, severe right leg weakness greater than arm/face weakness, and urinary incontinence. Which artery is mainly occluded?

Correct answer: E. Left ACA

Leg-predominant weakness plus urinary incontinence and apathy localizes to the paracentral lobule / medial frontal cortex — ACA territory — and since the weakness is on the RIGHT, the occluded vessel must be the LEFT ACA (contralateral rule).

Common trap: “Right ACA” (D) is the single most common wrong answer — students correctly identify ACA territory but then forget to flip the laterality to contralateral.

Verified — corrected past-paper key (Q169)

Q59. A pituitary tumor typically causes which visual defect?

Correct answer: B. Optic chiasm lesion

A pituitary tumor classically compresses the optic chiasm from below/midline, producing bitemporal hemianopia by damaging the crossing nasal retinal fibers from both eyes.

Common trap: “Optic tract lesion” (C) causes a homonymous, not bitemporal, field defect — an easy point along the visual pathway to confuse with the chiasm itself.

Verified — corrected past-paper key (Q180)

Q60. Which of the following is NOT a criterion used in the ABCD2 score for stroke risk after a TIA?

Correct answer: A. Hyperlipidemia

The ABCD2 score (Age, Blood pressure, Clinical features, Duration, Diabetes) predicts short-term stroke risk after a TIA; hyperlipidemia, while a general vascular risk factor, is not one of its five scored components.

Common trap: Hyperlipidemia is such a well-known general stroke risk factor that students often assume it must be part of every stroke risk score, including ABCD2.

Verified — corrected past-paper key (Q186)

Q61. An embolus from the internal carotid artery is most likely to lodge in which branch?

Correct answer: A. Middle cerebral artery (MCA)

The MCA is the more direct, higher-flow continuation of internal carotid artery blood flow, making it the most common destination for emboli arising from ICA (e.g., carotid plaque) disease.

Common trap: PCA (E) is a plausible-sounding “large artery” distractor, but it typically arises from the vertebrobasilar system, not the ICA, so ICA emboli rarely lodge there.

Verified — corrected past-paper key (Q187)

Q62. The arterial supply of the primary visual cortex is the:

Correct answer: C. Calcarine artery (PCA branch)

The calcarine artery, a branch of the PCA, supplies the primary visual cortex along the calcarine sulcus.

Common trap: MCA (A) is a common wrong guess since it feels like the “default” cortical artery, but the occipital pole / visual cortex is PCA territory.

Verified — corrected past-paper key (Q189)

Lecture 29 — Clinical Correlations — 3.0 hr — 15 questions

Q63. A 37-year-old woman presents with fatigue, difficulty speaking, ptosis, and diplopia that worsens throughout the day. Sensory exam is normal. What is the diagnosis?

Correct answer: E. Myasthenia Gravis

Fatigue, dysarthria, ptosis, and diplopia that WORSEN through the day, with a normal sensory exam, is the classic myasthenia gravis presentation — a postsynaptic NMJ disorder driven by anti-acetylcholine-receptor antibodies.

Common trap: Guillain-Barré syndrome (B) also causes weakness but is typically ascending and areflexic, and does NOT fluctuate through the day the way MG's fatigability does.

Verified — corrected past-paper key (Q190)

Q64. A 50-year-old man has ptosis, anhidrosis, and miosis due to a lung apex (Pancoast) tumor. Which muscle is responsible for the drooping eyelid?

Correct answer: D. Superior tarsal muscle

The ptosis of Horner's syndrome results from paralysis of the superior tarsal (Müller's) muscle — a smooth muscle with sympathetic innervation that provides accessory lid elevation alongside the CN III-innervated levator palpebrae superioris.

Common trap: Pupillary dilator (C) is responsible for the MIOSIS component of Horner's, not the ptosis — don't let two findings from the same syndrome get cross-wired.

Reclassified — corrected past-paper key (Q191), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative clinical-localization vignette.

Q65. The same 50-year-old man with a lung apex tumor also has miosis. Which muscle correlates with this finding?

Correct answer: C. Pupillary dilator

Miosis in Horner's syndrome results from loss of sympathetic tone to the pupillary dilator muscle, leaving the unopposed parasympathetic sphincter to constrict the pupil.

Common trap: Arrector pili (B) relates instead to the anhidrosis/flushing component of Horner's (loss of piloerection), not pupil size.

Reclassified — corrected past-paper key (Q192), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative clinical-localization vignette.

Q66. A 30-year-old woman with a UTI has urinary urgency. Which receptor mediates detrusor muscle contraction during voiding?

Correct answer: E. M3

Detrusor contraction for voiding is driven by parasympathetic input acting on M3 muscarinic receptors directly on bladder smooth muscle.

Common trap: Nicotinic (N, option C) mediates the autonomic ganglionic synapse, not the final effector action on the bladder wall itself — a common “wrong synapse” trap.

Reclassified — corrected past-paper key (Q194), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative clinical vignette.

Q67. A lesion in the left cerebral cortex affects the corticobulbar fibers. Which finding is true for this lesion?

Correct answer: C. Right weakness closing the mouth tightly

A left corticobulbar (UMN) lesion causes contralateral (right) LOWER facial weakness only — the upper face (eyelid closure, forehead) is spared because it receives bilateral cortical innervation, so the actual exam finding is weakness closing the mouth tightly, not eyelid closure.

Common trap: “Inability to close the right eyelid” (B) describes exactly what should be SPARED in a UMN facial lesion — that finding instead suggests a LMN (peripheral CN VII) lesion.

Reclassified — corrected past-paper key (Q203), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative UMN-localization vignette.

Q68. The mechanism of action for diazepam (a benzodiazepine) used in status epilepticus and anxiety disorders is:

Correct answer: C. Increases the frequency of Cl⁻ channel opening at the GABA-A receptor

Benzodiazepines like diazepam act allosterically on the GABA-A receptor to increase the FREQUENCY of chloride channel opening (only in the presence of GABA) — this is the key mechanistic distinction from barbiturates, which instead increase the DURATION of channel opening.

Common trap: Option B describes the barbiturate mechanism, not the benzodiazepine one — mixing up “frequency” (benzodiazepines) versus “duration” (barbiturates) is one of the most commonly tested pharmacology distinctions in this drug class.

Reclassified — corrected past-paper key (Q215), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as clinically-relevant pharmacology. (This item replaces an earlier internal-capsule question that the source archive itself had flagged as genuinely ambiguous; that flag was not being carried forward faithfully, so it has been swapped for this unambiguous item instead.)

Q69. A patient has vertical diplopia and difficulty looking down with the left eye, especially when the eye is adducted. What is the cranial nerve lesion?

Correct answer: A. Left CN IV

Difficulty looking down specifically when the eye is adducted, with vertical diplopia, is the classic presentation of a CN IV (trochlear nerve) palsy affecting the superior oblique muscle.

Common trap: CN III palsy (B) also affects several eye movements, but would additionally cause ptosis and a “down-and-out” resting position, not this isolated adduction-dependent downgaze deficit.

Reclassified — corrected past-paper key (Q206), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as a cranial-nerve exam vignette.

Q70. A patient has contralateral sensory loss and ipsilateral CN III palsy. This combination is caused by occlusion of the:

Correct answer: C. PCA

This combination — contralateral sensory loss (thalamic involvement) plus ipsilateral CN III palsy (midbrain cerebral peduncle/nerve involvement) — is a midbrain-thalamic (Weber-like) syndrome, since the PCA supplies both the thalamus and the midbrain.

Common trap: Basilar artery (E) is a tempting choice for any brainstem finding, but the co-occurrence of thalamic sensory loss specifically points to PCA rather than the basilar artery's more pontine territory.

Reclassified — corrected past-paper key (Q182), from the Lecture 28 pool; reassigned to Lecture 29 as an integrative brainstem/thalamic localization vignette.

Q71. Using the standard mnemonic for deep tendon reflex root levels (counting upward from the ankle: 1-2, 3-4, 5-6, 7-8), which nerve root level corresponds to the triceps reflex?

Correct answer: E. C7, C8

The mnemonic runs Achilles (S1,2) → patellar/quadriceps (L3,4) → biceps/brachioradialis (C5,6) → triceps (C7,8), moving from the most caudal reflex (ankle) to the most rostral (triceps).

Common trap: Biceps and triceps reflexes are frequently swapped — remember biceps (C5,6) is tested with the arm flexed, while triceps (C7,8) sits one segment “higher” in the numeric mnemonic, not lower.

Original — Neuro Exam (EP2) reflex-testing content. †

Q72. Which clinical finding, if present, favors a lower motor neuron (LMN) lesion over an upper motor neuron (UMN) lesion?

Correct answer: D. Fasciculations

Fasciculations — visible, spontaneous twitching of individual motor units — reflect denervation hypersensitivity and are a LMN finding; increased tone, hyperreflexia, Babinski sign, and clonus are all classic UMN findings.

Common trap: Because every other option is so strongly tied to “abnormal reflex/tone,” it is easy to assume ANY exam abnormality is UMN — fasciculations are the one listed sign that flips the localization to LMN.

Original — Neuro Exam (EP2) UMN/LMN comparison content. †

Q73. A patient has purely motor weakness with no sensory complaints. Examination shows fatigability that worsens through the day, but deep tendon reflexes remain preserved. This pattern best localizes to the level of the:

Correct answer: D. Neuromuscular junction

NMJ disorders (e.g., myasthenia gravis) classically present with purely motor findings, no sensory involvement, prominent fatigability, and — importantly — preserved reflexes, since the reflex arc itself remains structurally intact.

Common trap: Muscle (myopathic) disease also spares sensation and often spares reflexes, but myopathy classically causes fixed, proximal-predominant weakness without the fluctuating fatigability that is the hallmark of an NMJ disorder.

Original — Neuro Exam (EP2) localization-of-weakness content. †

Q74. A patient develops sudden bilateral leg weakness and loss of pain/temperature sensation below the umbilicus after an episode of severe hypotension, but vibration and proprioception remain intact. This is most consistent with:

Correct answer: B. Anterior cord syndrome

Anterior cord syndrome (classically from anterior spinal artery ischemia) affects the corticospinal and spinothalamic tracts bilaterally — causing motor and pain/temperature loss — while sparing the dorsal columns, so vibration and proprioception remain intact, exactly as described.

Common trap: Subacute combined degeneration (D, from B12 deficiency) causes the mirror-image pattern — dorsal column loss (vibration/proprioception) with corticospinal involvement, but pain/temperature classically spared — an easy pattern to invert under pressure.

Original — Neuro Exam (EP2) spinal cord syndrome content. †

Q75. A patient asked to perform rapid alternating pronation-supination of the forearm performs the movements slowly and irregularly. This finding is best termed:

Correct answer: C. Dysdiadochokinesia

Dysdiadochokinesia is specifically the inability to perform rapid alternating (agonist-antagonist) movements, such as rapid pronation-supination — a classic cerebellar hemisphere sign.

Common trap: *Dysmetria (A) refers to errors in judging distance/endpoint (overshoot or undershoot) during a targeted movement, such as finger-to-nose testing — a related but distinct cerebellar sign that is often confused with dysdiadochokinesia.*

Original — Neuro Exam (EP2) cerebellar examination content. †

Q76. A large cerebellopontine (CP) angle tumor classically produces a nystagmus with a large-amplitude, slow component when looking toward the side of the lesion and a smaller, fast component when looking away. This is known as:

Correct answer: B. Bruns' nystagmus

Bruns' nystagmus is a dual-mechanism nystagmus classically seen with large CP angle tumors (e.g., vestibular schwannoma), combining a large-amplitude vestibular nystagmus toward the lesion side with a smaller cerebellar gaze-evoked nystagmus away from it.

Common trap: *Downbeat nystagmus (A) is a distinct finding classically associated with the craniocervical junction (e.g., Chiari malformation), not a CP-angle-tumor sign — don't let "another named nystagmus" substitute for the specific CP-angle pattern.*

Original — Neuro Exam (EP2) cerebellar dysfunction content. †

Q77. Which clinical feature helps distinguish botulism from myasthenia gravis?

Correct answer: C. Botulism characteristically also involves autonomic dysfunction, while myasthenia gravis does not

Botulism (a presynaptic NMJ disorder caused by blockade of acetylcholine release) characteristically produces autonomic involvement (e.g., dilated pupils, dry mouth, ileus) alongside descending flaccid paralysis, whereas myasthenia gravis (postsynaptic, anti-AChR-antibody-mediated) classically spares autonomic function.

Common trap: *Option D reverses the actual synaptic mechanisms — botulism is presynaptic (like Lambert-Eaton) and myasthenia gravis is postsynaptic; both disorders can cause ptosis/diplopia (ruling out B), and statement A, while true for both, does not distinguish them from each other.*

Original — Neuro Exam (EP2) NMJ disease content. †
